Project Submission–Database Administration

STUDENT EXAMINATION DATABASE

STUDENT DETAILS

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Lastdateof submission	25/09/2026
Github link	

1. Pro blem Statement & Objectives – 5 Marks

Problemstatement

Createa database for maintaining examination details, subjects, students, marks and results. The database should organize student and subject information, store examination records and marks, calculate grades, and generate a student

Objectives

• M•aintain student details in a structured database.A vehicle service center needs a database to efficiently
• Store and manage subject details.
• Maintain examination records for students and subjects information about customers, their vehicles, and the services performed on
• Store marks and calculate grades automatically.
those vehicles. Maintaining these details manually makes it difficult to track
• Provide easy insertion, updating, deletion and retrieval of records.
service history, retrieve customer information, and calculate service-related
• Generate student result reports using SQL queries and joins.
• Maintain data accuracy using primary keys, foreign keys, NOT NULL and UNIQUE constraints.costs. The Vehicle Service Management Datbase stores this information in related tables and allows the service center to perform data insertion, updating, deletion, searching, filteri

2. DatabaseDesign –10Marks

2.1 Database Structure

TheStudent Examination Database is designed to store student,subject,examination,marksand result information in related tables. Primary keys uniquely identify records and foreign keys establish relationships between the tables

Relationships

- One student can have many examination/marksrecords.
- One subject can appear in many examination/marks records.
- One examination can contain marks for many students and subjects.
- Each marks record can be used to generate a corresponding grade/result.

Table used

Table name	Purpose
Student	Stores student ID, name, department and year
Subject	Stores subject ID, subject name, maximum marks and credits.
Exam ination	Stores examination ID, examination name and examination date.
Marks	Stores student marks for each subject and examination
Result	Stores grade and result status calculated from marks

2.2 Relationship Summary

From	Relationship	To
Student	1:M	Marks
Subject	1:M	Marks
Examination	1:M	Marks
Marks	1:1	Result

2.2 Entity Relationship Diagram

The ER diagramshowsthemaintables,attributes,primary keys, foreign keys and relationships used in the Student Examination Database.

STUDENT	MARKS	SUBJECT	EXAMINATION	RESULT
StudentID (PK)	MarkID (PK)	SubjectID (PK)	ExamID (PK)	ResultID (PK)
StudentName	StudentID (FK)	SubjectName	ExamName	MarkID (FK)
Department	SubjectID (FK)	MaxMarks	ExamDate	Grade
Year	ExamID (FK)	Credits		ResultStatus

2.3 Technologies Used

Technology / Tool	Purpose
MySQL	Database management system used to store and manage student, subject, examination, marks and result record
MySQL Workbench	Used to create tables, execute SQL queries, manage the database and view query results.
SQL	Used for database creation, data manipulation, filtering, grouping, joins and result calculation.

3. Practical Implementation –15Marks

3.1 Database and Table Creation

DatabaseCreation

```
CREATE
DATABASEstudent_exam_db;

USE
student_exam_db;
```

Student Table

```
Stude
        ntID
  INT PRIMARY
SKtEuYd,entName VARCHAR(100)
NOT NULL,
Department VARCHAR(50)
NOT NULL,
YearINT NOT
NULL );
```

Subject Table

```
CREATE TABLE Subject ( SubjectID
INT PRIMARY KEY, SubjectName
VARCHAR(100) NOT NULL
, MaxMarks INT NOT NULL,
Credits INT );
```

3.2 Constraints and Relationships

Thedatabaseusesprimarykeystouniquelyidentifyrecords,fore ignkeystoconnectrelatedtables, NOT NULL to ensure required values,and UNIQUEtoprevent duplicate values where necessary.

Table	Primary Key	Foreign Key	Other Constraints
Student	StudentStudent	-	NOT NULL
Subject	id Exam id	-	NOT NULL
Examination		-	NOT NULL
Marks	Mark id	StudentID, SubjectID, ExamID	NOT NULL
Result	Result id	Mark id	NOT NULL

Marks Table

```
CREATETABLEMarks (
MarkIDINT PRIMARY KEY,
  StudentID INT NOT NULL,
ExamID INT NOT NULL,
Marks INT NOT NULL,
FOREIGN KEY (StudentID) REFERENCEStudent(StudentID),
FOREIGN KEY (Subje ctID) SREFERENCESubject(SubjectID),
FOREIGN KEY (ExamID)REFSERENCESExamination(ExamID)
);
```

Result Table

```
ResultID INT PRIMARY KEY,
MarkID INT NOT NULL,
Grade VA)RC;HAR(5),
ResultStatus VARCHAR(20),

FOREIGNKEY REFERENCE Marks(MarkID)
(MarkID) ); S
```

-

3.3 Data Insertion

Stude ntData

```
INSERTINTOstudent VALUES
(1,'Arun Kumar','BCA',3),
((3 2 , ' , R ' p a v r i i S y h a a d n k e a V r ' , i ' B , ' c b A c ' , 3 a ) ' , , 3 ) ,
(4,’meena devi’,’bca’,3)
```

Subj ect Data

```
INSERTINTOSu bject VALUES
(101,'Database Management Systems',100,4),
(102,'PythonProgramming',100,4),
(103,'Web Technology',100,4),
(104,'Software Engineering',100,4);
```

ExaminationData

```
IN SERTINTOExamination VALUES
1001 , ‘semester examination ‘,2026-08-20’);
```

Marks data

```
INSERTINTOMarks
VALUES (1,1,101,1001,86),
(2,1,102,1001,91),
(3,1,103,1001,88),
(4,1,104,1001,87),
(5,2,101,1001,78),
(6,2,102,1001,82),
(7,2,103,1001,76),
 (8,2,104,1001,82);
```

3.4 SQL operations

INSERT

```
INSERT INTO student ALUES (5,’kartik raj’, ‘BCA’,3);
```

SELECT

```
SELECT * FROM Student;
```

UPDATE

```
UPDATE Student SET Department='BCA' WHERE StudentID=5;
```

DELETE

```
DELETE FROM Marks WHERE MarkID=12;
```

WHERE / ORDERBY

```
OSERLEDCTE *R F BROYM Marks WHERE Marks > 80;
SELECT * FROM Marks ORDER BY Marks DESC;
```

LIKE / BETWEEN

```
SELECT * FROM Student WHERE StudentName LIKE 'A%';
SEL ECT * FROM Marks WHERE Marks BETWEEN 60 AND 90;
```

Theseoperationsare usedtoinsert,retrieve,modify,remove,filterand sort examination data.

3.5 GradeCalculation and Student Result Report

Grade Calculation

```
SELECTStudentID, SubjectID,Marks,
CASE
WHEN Marks >= 90 THEN 'A+'
WHEN Marks >= 80 THEN 'A'
WHEN Marks >= 70 THEN 'B'
WHEN Marks >= 60 THEN 'C'
ELSE 'F' END AS Grade
FROM Marks;
```

GROUP BY / Average

```
SELECTStudentID,COUNT(*)AS Subjects, AVG(Marks) AS AverageMarks FROM Marks GROUP BY StudentID;
```

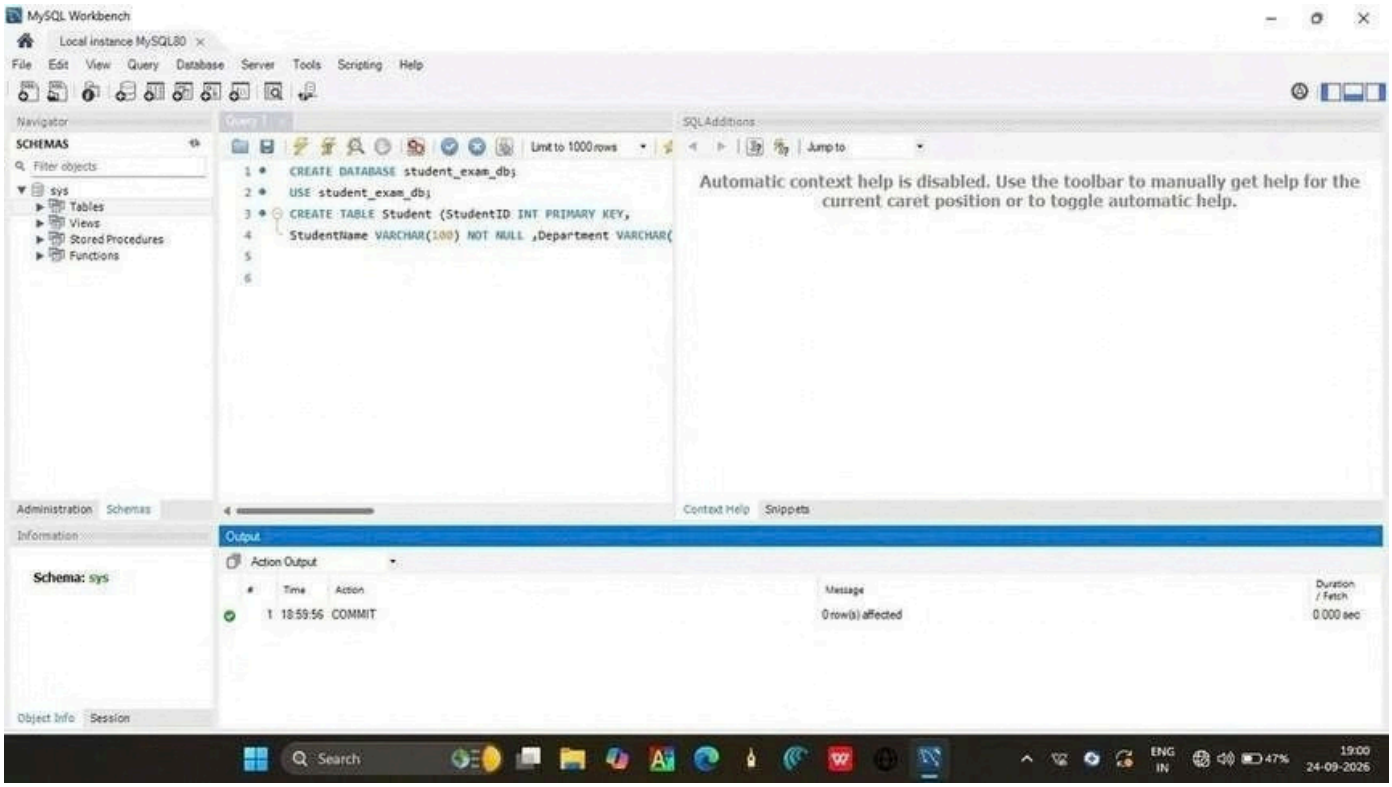
Student Result Report

```
SELECT s.StudentName, sub.SubjectName, m.Marks,r.Grade,r.ResultStatus
FROM Student s
FROM Customer c INNER JOIN Marks m ON s.StudentID = m.StudentID INNER JOIN Subject
sub ON m.SubjectID = sub.SubjectID
INNER JOIN Vehicle v ON c.customer_id = v.customer_id
INNER JOIN Result r ON m.MarkID = r.MarkID;
```

The final report combines student, subject, marks and grade information so that the result can be viewed in one report

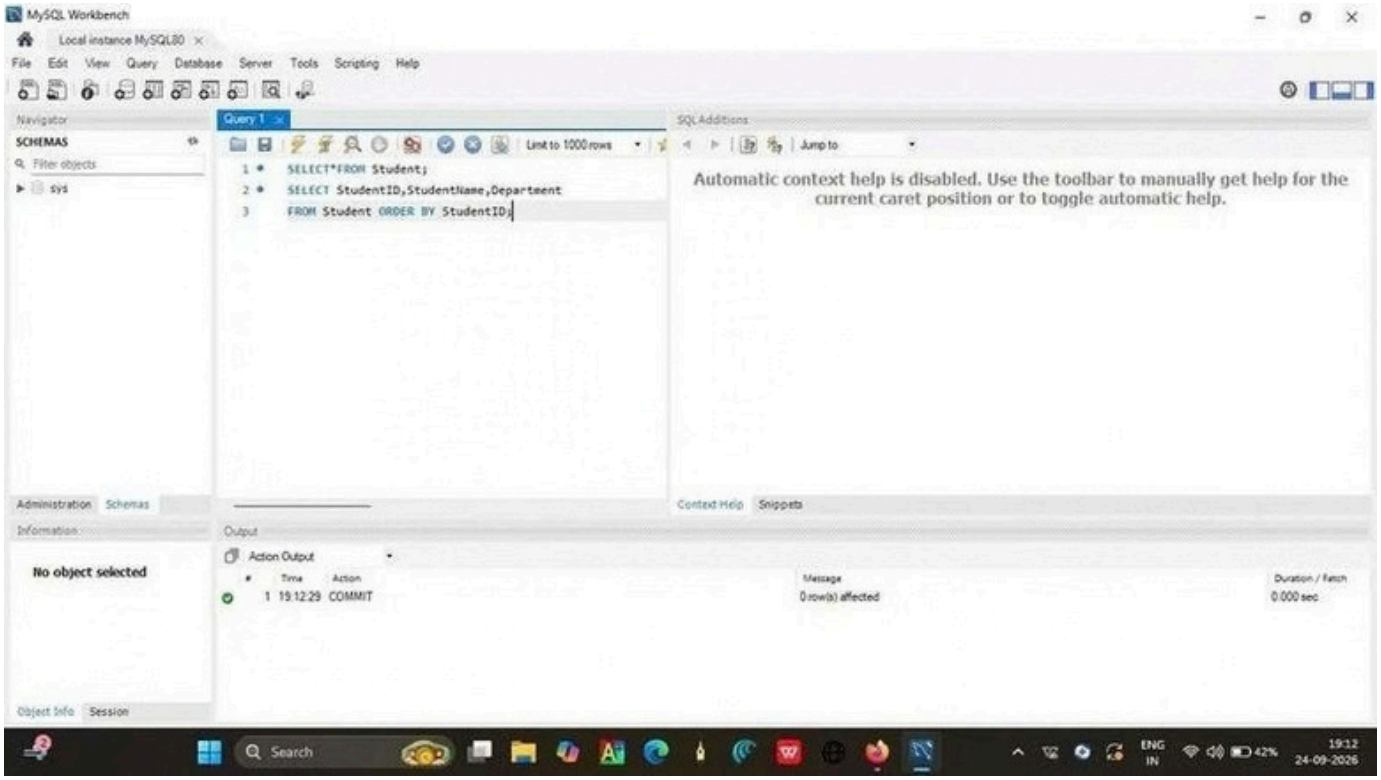
4. Output Screenshots–10Marks

Screenshot1–DatabaseandTables



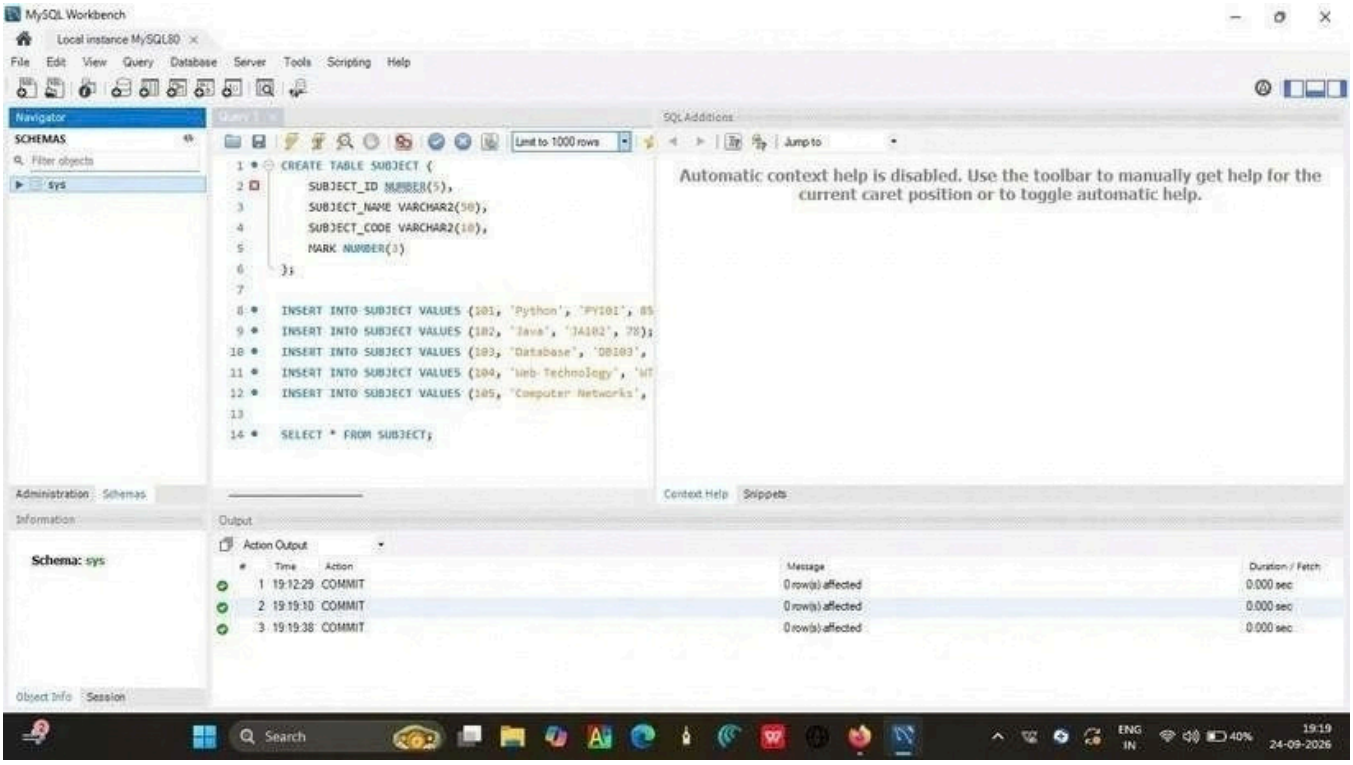
4. Output Screenshots – 10 Marks

Screenshot 2 – Student Table with Records



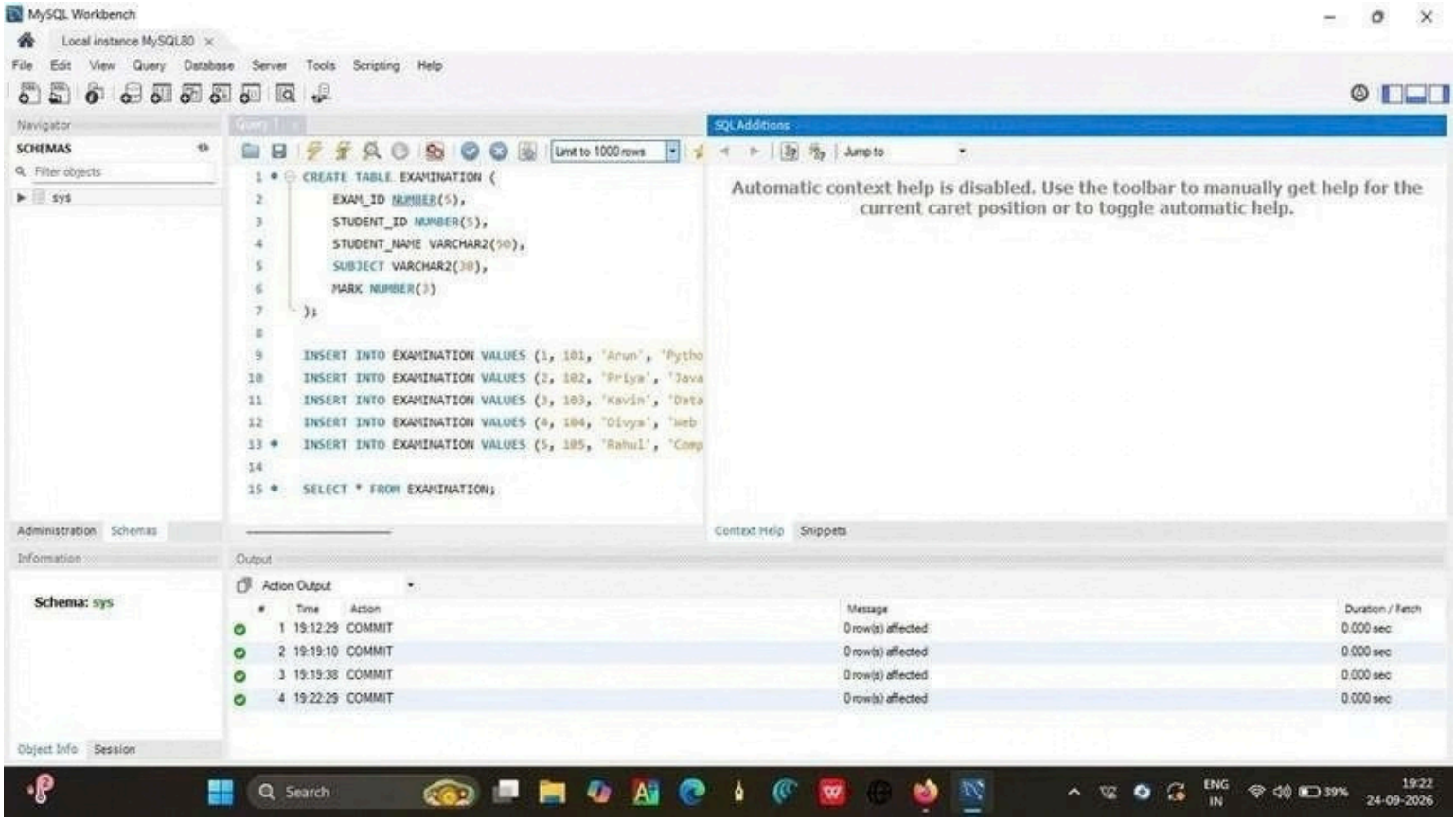
4.Output Screenshots-10 marks

Screenshot 3 – Subject Table with Records



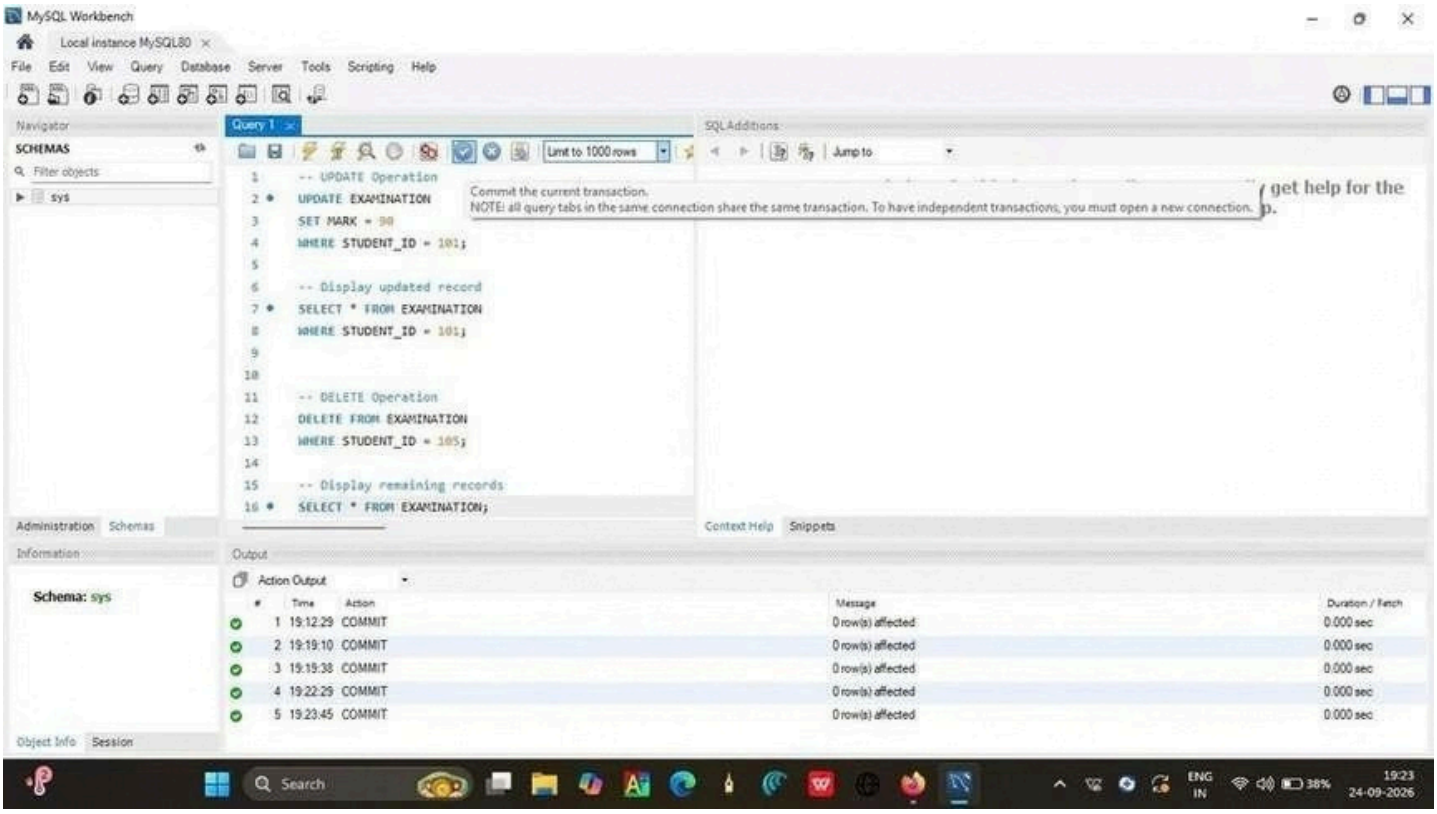
4.Output Screenshots - 10 Marks

Screenshot 4 – Examination and Marks Records



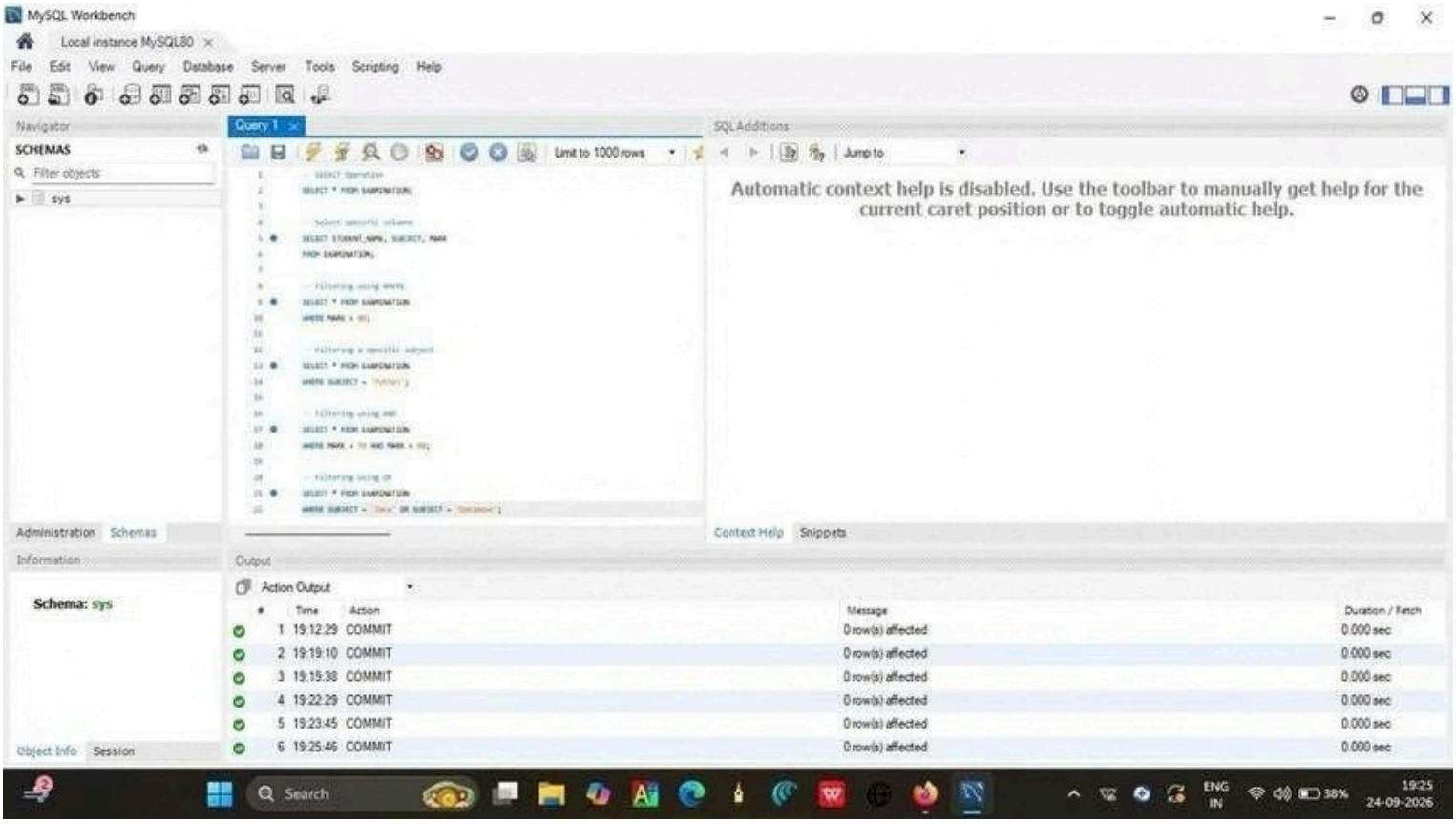
4.Output screenshots - 10 Marks

Screenshot 5 – UPDATE/DELETE Operations



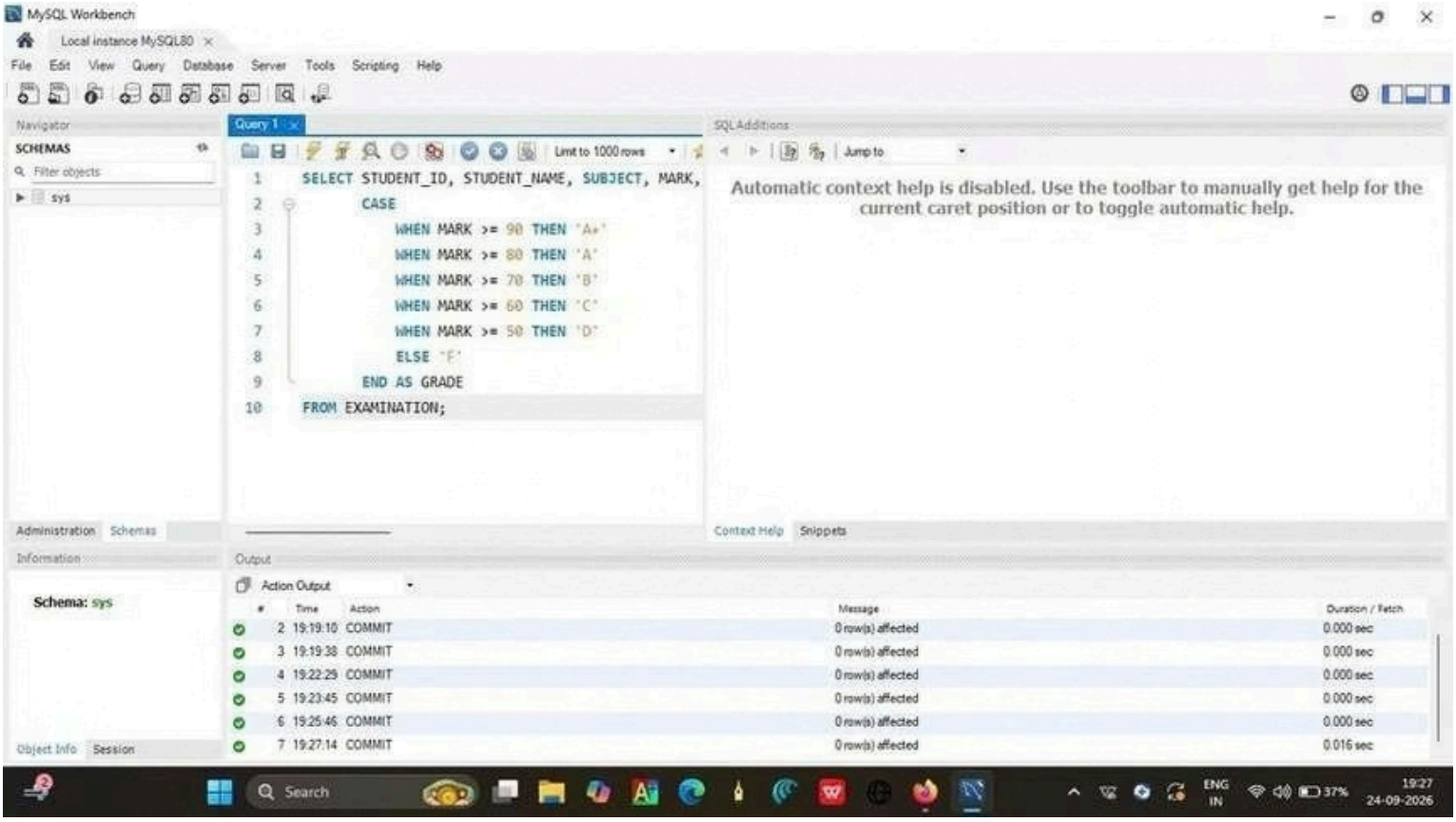
4. Output Screenshots – 10 Marks

Screenshot 6–SELECTandFilteringOperations



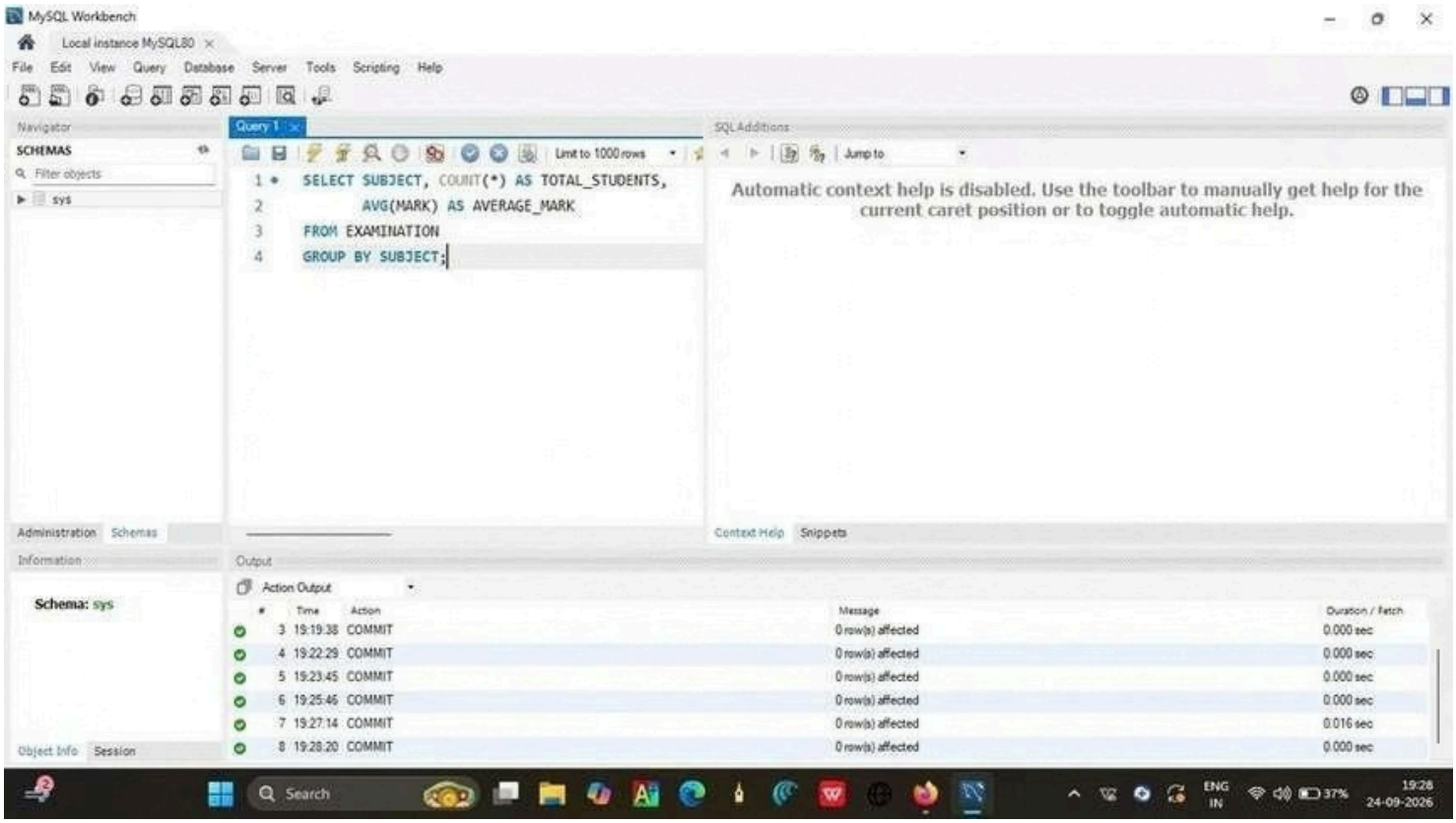
4. Output Screenshots – 10 Marks

Screenshot 7 – Grade Calculation



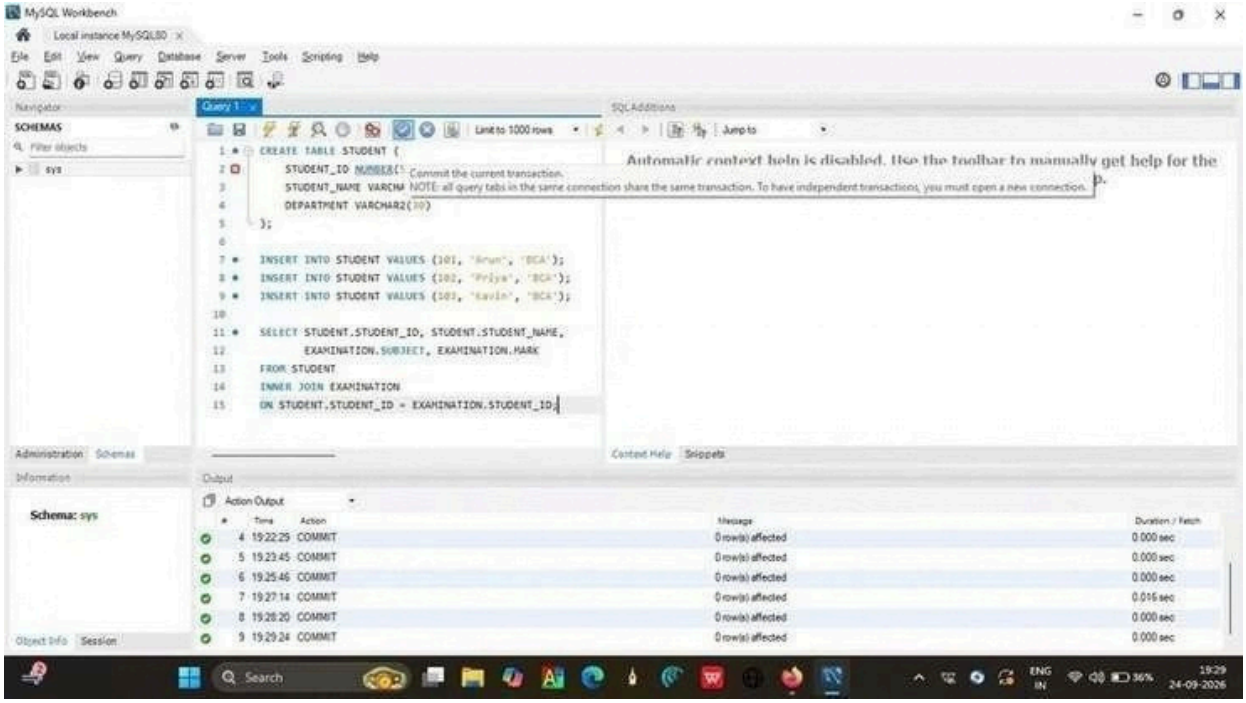
4. Output Screenshots – 10 Marks

Screenshot8 –GROUP BY Operation

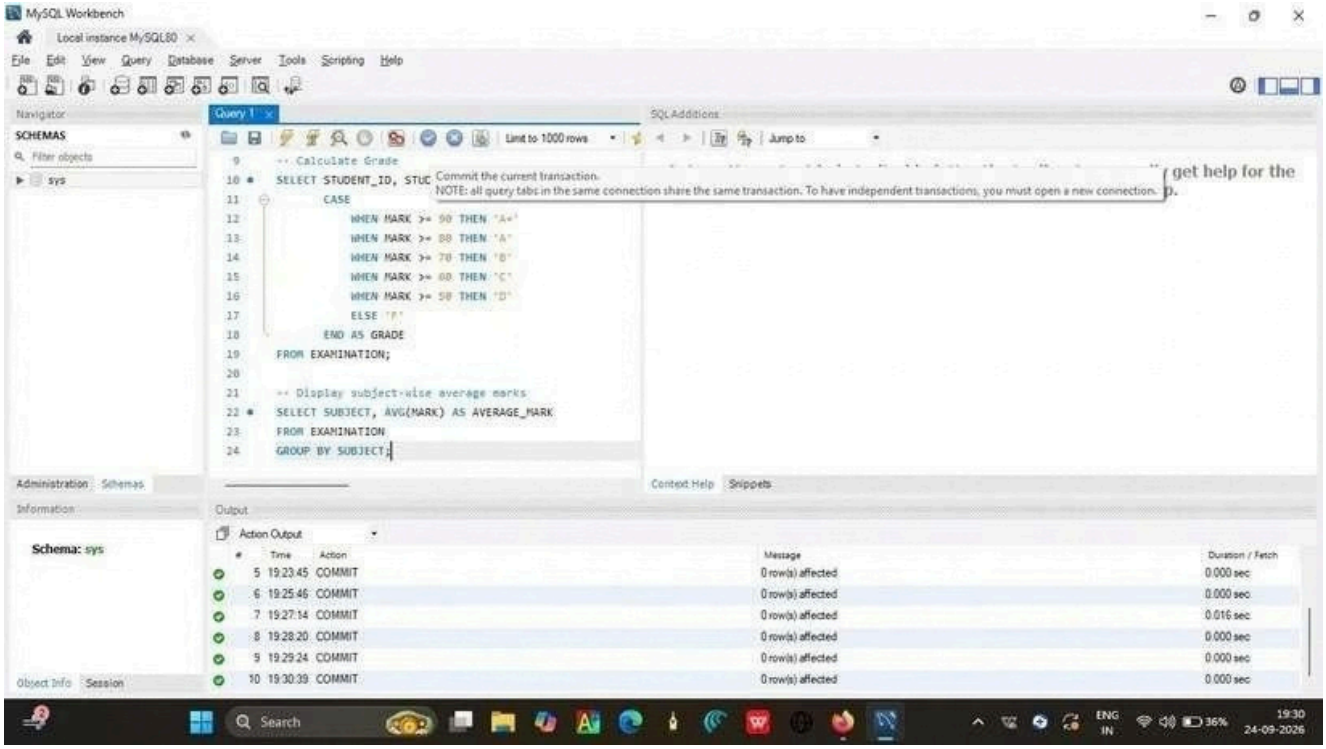


4.Output Screenshots -10 marks

Screenshot9–INNERJOINOperation



Screenshot10–FinalDatabase/QueryResults



5. Technical Explanation – 5 Marks

Data Retrieval

WHEREfiltersrecordsbasedonacondition,ORDERBYsortsrecords,LIKE searches for partial values, BETWEENretrievesvalueswithinarange,andGROUPBYgroupsrecordsaccording to a selected column.

SQL Operations

INSERT adds records, SELECT retrieves records, UPDATE modifies existing records, and DELETE removes records. These operations are used to manage the database.

Database design

The database is divided into Student, Subject, Examination, Marks and Result tables to keep the data organized and reduce unnecessary duplication. Foreign keys connect the related tables.

Keys and Constraints

Primary keysuniquelyidentify records. Foreign keys establish relationships between tables. NOT NULL ensures required values are entered, while UNIQUE prevents duplicate values

Aggregate Functions and Joins

COUNT, SUM, AVG, MAX, and MIN perform calculations on groups of records. INNER JOIN combines related information from Customer, Vehicle, and Service tables using their keys.

6.Conclusion - 5 Marks

Future Enhancements

- Student login and authentication
- Onlin eexaminationschedulingandnotifications.
- Automated resultpublication.
- Printablem arksheetsandresultreports.
- Attendance integration.
- Faculty/admindashboardforexaminationmanagement.
- Integration withawebormobileapplication.
- Automated databasebackupandrecovery.

Conclusion

T he student Examination database provides a stuctural way to manage student details,subject,examination records marks and results.Primarykeys, foreignkeys, constraints, SQL operations,filtering,grouping,gradecalculation and INNERJOIN were used to implement the required database functions.Theproject demonstrates practica lunderstanding of relational database design and SQL-based examination data management.